



**S. B. JAIN INSTITUTE OF TECHNOLOGY,
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Practical No. 10

Aim: Apply prompt engineering to construct a prompt for learning the concept of Adversarial Search.

Name of Student:

Roll No.:

Semester/Year: VI/III

Academic Session:

Date of Performance:

Date of Submission:

AIM: To design and refine an effective prompt using prompt engineering techniques for learning the concept of Adversarial Search from Artificial Intelligence: A Modern Approach.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

The objectives and expected learning outcome of this practical are:

- To be able to understand the concept of a *prompt* (input given to an AI model).
- To be able to understand the concept of *prompt engineering* (designing inputs to get desired outputs)
- To be able to work with basic idea and models like ChatGPT.
- To be able to distinguish between a vague prompt and a well-structured one.

THEORY:

i. Fundamentals of Prompt Engineering

Prompt engineering is the process of designing and structuring inputs (prompts) to get accurate, relevant, and well-formatted outputs from AI systems such as ChatGPT. Prompt engineering is the process of designing and structuring inputs (prompts) to get accurate, relevant, and well-formatted outputs from AI systems such as ChatGPT. Prompt engineering is how you ask a question to AI so that it gives the best possible answer.

ii. Characteristics of a Good Prompt

A good prompt is based on the following core principles:

- **Clarity** → Avoid ambiguity
- **Specificity** → Define exact task
- **Context** → Provide background
- **Constraints** → Limit output (length, format, style)
- **Intent** → Clearly state what is required

Example:

- Weak: “*Explain AI*”
- Strong: “*Explain AI in 5 bullet points with real-world examples for beginners*”

iii. Prompt Patterns / Techniques

The common patterns of the prompt along with their usage are as follows:

a) Role-Based Prompting

“You are an AI professor...”

b) Step-by-Step Prompting

“Explain step-by-step...”

c) Few-shot Prompting

Provide examples in the prompt

d) Instruction + Constraints

“Explain in 100 words using simple language”

e) Chain-of-Thought (Guided Reasoning)

“Show reasoning before final answer”

iv. Task-Oriented Prompting

Based on the tasks we can have different ways to provide the prompt. Following are the examples to design prompts for different tasks:

- **Learning/Explanation**
- **Coding**
- **Summarization**
- **Question generation**
- **Problem solving (AI, Cloud, etc.)**

v. Iterative Refinement (Very Important)

The first prompt may not be the best one. So, we must learn how the prompt can be refined to get most appropriate outcome:

- First prompt $\neq$ best prompt
- Improve by:
 - Adding constraints
 - Asking follow-up questions
 - Correcting output

Concept:

Prompt $\rightarrow$ Output $\rightarrow$ Refine $\rightarrow$ Better Output

vi. Evaluation of AI Output

It is very important to evaluate the output generated by AI. Critical thinking should be done to find the answer to the following:

- Is the answer **correct**?
- Is it **complete**?
- Is it **biased or vague**?
- Does it follow instructions?

vii. Ethical and Responsible Use

- Avoid Misuse and Plagiarism
 - Do not copy AI-generated content blindly for assignments or research
 - Always ensure originality and proper attribution when required
- Verify Information
 - AI tools like ChatGPT may generate incorrect or outdated information
 - Cross-check important facts before using them in academic or professional work
- Use AI as a Support Tool
 - AI should assist learning, not replace critical thinking
 - Students should understand concepts rather than depend entirely on AI-generated answers

viii. Basic Applications in Academics

- Generate notes
- Create MCQs and scenario-based questions
- Debug code
- Explain complex topics simply

ix. Perform the following tasks:

- Convert a bad prompt → good prompt
- Generate:
 - Notes
 - Quiz questions
 - Code snippets
- Design a prompt for:
 - Adversarial Search explanation

EXPECTED OUTPUT:

- Definition
- Game tree explanation
- Minimax working
- Alpha-Beta optimization
- Example
- Complexity analysis

INPUT & OUTPUT:

Design the prompts and provide the analysis using the following criteria:

Attempt	Prompt Quality	Output Quality	Improvements
1	Basic	Poor	Add clear structure, specify required topics
2	Improved	Moderate	Add constraints like format, steps, and word limit
3	Refined	High quality	Final optimized prompt, no major changes needed

Attempt 1 (Basic Prompt)

Prompt: Explain adversarial search

Output:

Adversarial Search (Short Explanation)

• Definition:

Adversarial search is a technique in AI used for decision-making in situations where multiple agents compete against each other (like games).

• Game Tree:

A tree structure where:

- Nodes = game states

- Edges = possible moves
 - Root = current state
 - Leaves = final outcomes (win/lose/draw)
 - **Minimax Algorithm:**
 - One player tries to **maximize** the score
 - Opponent tries to **minimize** the score
 - The algorithm chooses the best possible move assuming the opponent plays optimally
 - **Alpha-Beta Pruning:**
 - Optimization of minimax
 - Cuts off unnecessary branches
 - Reduces computation time
 - **Example:**
Chess, Tic-Tac-Toe
 - **Time Complexity:**
 $O(b^d)$
(b = branching factor, d = depth)
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In short:

Adversarial search helps AI make the best decision in competitive environments by predicting opponent moves.

Output Analysis:

- Very general
- Missing details
- No structure

Attempt 2 (Improved Prompt)

Prompt: Explain adversarial search with minimax and example

Output :

Adversarial Search with Minimax (Short Explanation)

• **Adversarial Search:**

Used in AI for decision-making in competitive situations where one player's gain is another player's loss (e.g., games).

Minimax Algorithm

- Works for two players:
 - **MAX player** → tries to maximize score
 - **MIN player** → tries to minimize score
- Process:
 1. Generate game tree
 2. Go to leaf nodes (end states)
 3. Assign values (win = +1, loss = -1, draw = 0)
 4. Backtrack:
 - MAX chooses highest value

- MIN chooses lowest value
- 5. Final decision = best move for MAX

Example (Tic-Tac-Toe)

- Suppose MAX is 'X' and MIN is 'O'
- At final states:
 - X wins $\rightarrow +1$
 - O wins $\rightarrow -1$
 - Draw $\rightarrow 0$

Backtracking:

- MIN tries to reduce score
- MAX selects move giving highest value

☞ So AI chooses the move that guarantees the best possible outcome.

Time Complexity

$O(b^d)$

(b = branching factor, d = depth)

Output Analysis:

- Better explanation
- Still lacks structure
- No constraints

Attempt 3 (Refined Prompt)

Prompt: You are an AI professor. Explain adversarial search in simple terms including:

- Definition
- Game tree
- Minimax algorithm
- Alpha-beta pruning
- Example
- Time complexity

Explain step-by-step in bullet points within 200 words.

Output :

Adversarial Search (Step-by-Step Explanation)

• Definition:

Adversarial search is used in AI for decision-making in competitive environments where two players oppose each other (one wins, one loses).

• **Game Tree:**

- A tree structure representing all possible moves
 - Root = current state
 - Nodes = game positions
 - Edges = possible moves
 - Leaves = final outcomes (win/lose/draw)
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• **Minimax Algorithm:**

- Two players: **MAX** (AI) and **MIN** (opponent)
 - Steps:
 1. Expand game tree to terminal states
 2. Assign values (win = +1, loss = -1, draw = 0)
 3. Backtrack values upward
 4. MAX chooses highest value
 5. MIN chooses lowest value
 - Final move = best guaranteed outcome
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• **Alpha-Beta Pruning:**

- Optimization of minimax
 - Skips branches that won't affect final decision
 - Reduces computation time
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• **Example:**

- Tic-Tac-Toe or Chess
 - AI evaluates moves and selects best one assuming opponent plays optimally
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• **Time Complexity:**

- Minimax: $O(b^d)$
- Alpha-Beta: $O(b^{(d/2)})$ (best case)

Output Quality:

- Clear and structured
- Complete answer
- Easy to understand

CONCLUSION: Hence, we have successfully applied prompt engineering to construct a prompt for learning the concept of Adversarial Search.

DISCUSSION QUESTIONS:

1. What is prompt engineering?

2. Why is prompt refinement important?

3. What is Alpha-Beta pruning?

4. Difference between weak and strong prompts?

REFERENCES:

- <https://learnpromptengineering.dev>
- <https://www.linkedin.com/pulse/40-prompt-engineering-resources-upskilling-spencer-taylor-xje0e/>
- <https://justprompting.in/>